

Introduction

Our Census TOP Data Landscape and Integration team primarily explored some datasets from the U.S. Census Bureau during the project phase, focusing on the accessibility, usability, and integration of these census datasets for large language modeling (LLM) systems. We identified and analyzed several representative, frequently used datasets. Our goal was not only to analyze each dataset individually but also to uncover broader data contexts and situations, identify inconsistencies between different products, and identify potential barriers users or large language models might face when using the data. Throughout the project, the support team analyzed and studied datasets such as the American Community Survey (ACS), economic census, population estimates, health insurance estimates (SAHIE), international trade data, and higher education employment outcomes (PSEO). These are commonly used datasets, and the final research results revealed the importance of building tools capable of interacting with federal open data.

Critical Analysis

To evaluate our project, I believe its main strength lies in the diversity of the datasets we explored. Our team examined multiple different aspects of the data. However, we found variations in the census data in terms of format, time span, and geographic detail. Census data is not a strictly uniform database but rather comprises numerous, dispersed datasets, each with its own update frequency, geographic reference, and format. We constructed a unified data environment through data sifting and cleaning and analyzed these datasets to explore the meaning of the data. We found significant data gaps in the U.S. Community Survey (ACS) and PSEO datasets, such as some major states not participating in national data collection at all,

which reduces the national representativeness and accuracy of the data. Therefore, one issue we identified is how LLM communicates uncertainty to users when analyzing queries containing incomplete data. In-depth analysis of the team's results indicates that while each team member successfully cleaned and analyzed data in their chosen areas, our project currently resembles an independent analysis of each dataset rather than a fully integrated whole, lacking a comprehensive inventory of all data and cross-dataset consolidation analysis.

Assessment of Methodological Choices

In this project, the team's methodology choice was to use the API to obtain raw data rather than manually selecting and downloading static CSV files from the online dataset. This approach effectively proves the accessibility and availability of the census data API. This suggests that LLM applications may also be able to access data directly via the API when they need to retrieve datasets included in the census. Furthermore, data cleaning methods played a crucial role in our analysis. The lack of standardized variables and formats in this data would have been a significant obstacle to data integration in an LLM system. By transforming numerical variables, removing duplicates, and creating new, complete, and clean lists, this process demonstrated the dataset screening techniques that future AI agents would face, elevating the analysis from theoretical data availability to practical technical feasibility.

Implications for the Project

The most important finding is that many datasets suffer from inconsistent variable formatting and obscure encoding schemes. Many API responses return coded numbers or codes rather than human-readable labels. This means the integration process for Large Language

Models (LLMs) need improvement, requiring the introduction of intermediaries to translate user-input language commands into Census Bureau-compatible code, enabling direct text-to-API navigation and helping LLM query URLs more efficiently. Furthermore, identifying data gaps during the project has a significant impact on the reliability and trustworthiness of the final product. Incomplete or unevenly covered datasets mean LLMs must be able to generate responses that address uncertainties or concerns when data is incomplete or missing from certain states, to ensure users don't mistake partial data for complete national data.

Suggestions for Additional Research

Based on the issues raised above, the following directions will help strengthen and improve this project. First, regarding cross-dataset integration, a platform could be developed to fix the structure of census data, such as geographic regions and variable types, ensuring a unified format for the datasets. This model can fix inconsistent datasets, enabling the LLM to understand the relationships between datasets and aggregate data, preventing variable mismatches. This will make integration more concrete and help to design a practical MCP workflow. Another proposal is to create a universal codebase capable of collecting variables and labels from census documents, helping people understand the meaning of code numbers appearing in API retrieval processes, ensuring both the accuracy of instructions and the LLM's understanding and retrieval of data. This ensures a balance between AI software and human intervention in the final product.

Analysis of the Census Bureau's data revealed that while LLM has significant potential, it is currently constrained by severe structural disagreements and technical barriers. Our investigation of the datasets we identified indicates that census data is not a single-format database, but rather a collection of independent datasets, each with its unique update frequency,

geographic location, and coding standards. While our team successfully unified and filtered these different domains, the findings confirm that the current ecosystem resembles a collection of independent datasets rather than a fully integrated whole. Therefore, standardizing the format is crucial when enabling LLM to use the data. This project also highlighted the gap between LLM data numbering and users, which requires standardized coding to help both LLM and human users understand it. By continuously improving the numbering system and focusing on integrating and standardizing the format of census data, LLM can more easily facilitate data access, extraction, and analysis for a wider range of users.